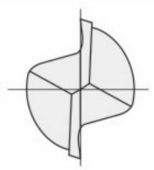
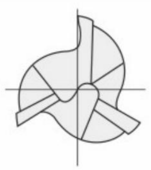
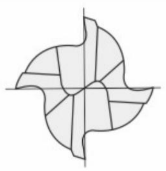
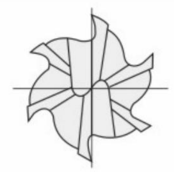
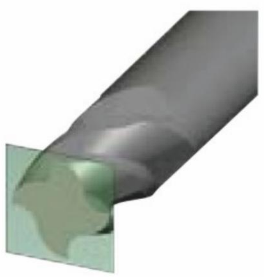
## 铣刀的刃数 Number Of Flute

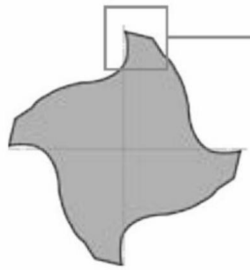
根据铣刀用途选择刃数非常重要。下面对各种刃数的优缺点及其用途进行介绍。
It is important to choose the number of flute of the end mill according to the usage. Introduction of the advantages,disadvantages, and usage of each number of flute are as below.

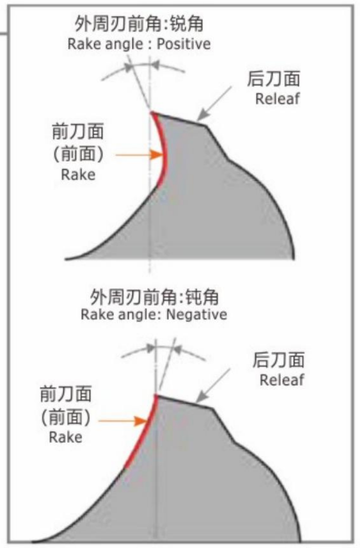
| | |  2刃 2-Flute |  3刃 3-Flute |  4刃 4-Flute |  6刃 6-Flute |
|---|---|---|---|---|---|
| 特点 Features | 优点 Advantage | 排屑良好 Well chip removal | 排屑良好 Well chip removal | 刀具刚性高 High tool rigidity | 刀具刚性高 High tool rigidity |
| | 缺点 Disadvantage | 刀具刚性低 Low tool rigidity | 难以测量外径 Difficult to measure dia. | 排屑不畅 Low chip removal | 排屑不畅 Low chip removal |
| 用途 Usage | | 通用 For General use | 沟槽和侧面 Slotting & Side milling | 侧面 Side milling | 侧面 Side milling |

## 铣刀的前角 Rake angle at peripheral cutting edge

铣刀的前角因用途而异，主要取决于加工材料
Variety of rake angles to support different cutting materials.








前角为锐角造型的铣刀注重锋利度,一般适用于非铁金属和低硬度的加工材料。前角为钝角造型的铣刀具有高强度刀刃,适用于高硬度钢等材料。
End mill with positive rake angle has a sharp flute that generally compatible with nonferrous metals and low hard materials. On the other hand, the negative rake angle has strong cutting edge to support high hard materials machining.





## 切削速度、主轴转速、进给速度的计算方法 Calculation for Cutting Speed, Spindle Speed and Feed

| | | | |
|---|---|---|---|
| 切削速度 Cutting Speed | VC (m/min) | = 丌（圆周率3.14）× D（外径mm）× n（主轴转速$min^{-1}$）/ 1,000 | |
| 主轴转速 Spindle Speed | n ($min^{-1}$) | = VC÷丌÷D ×1,000 | |
| 进给速度 Feed | vf (mm/min) | = n × fz × Z（刃数） | |
| 每刃进给量 Feed per Tooth | fz (mm/tooth) | = Vf / n × Z（刃数） | |
| 切削排出量 Metal Removal Rate | Q ($cm^3$/min) | = ap× ae × vf / 1,000 | ap：轴向深切量（mm）Axial depth of cut<br>ae：径向深切量（mm）Radial depth of cut<br>vf：进给速度（mm/min）Feed |
| 理论表明粗糙度 ※平坦部 Theoretical Surface Finish ※Plane | h (μm) | = ae2 × 1,000 ÷ 8R | ae：径向深切量（mm）Radial depth of cut<br>R：球头半径（mm）Feed |

### 使用的设备主轴不满足切削参数表的转速时…

利用下述的计算式以相同的比率降低主轴转速和进给速度后使用。
When maximum speed of the machine spindle less than value of recommended milling conditions, adjust conditions by calculation as follows.

- 根据所使用的主轴转速[n]与切削参数表的主轴转速[n1]计算比率[α]。
  Rate (α)is calculated by chosen Spindle Speed (n) and Recommended Spindle Speed (n1).

$$n(\min^{-1}) \div n1(\min^{-1}) = \alpha$$

比例
Rate

- 将上述比率[α]乘以切削参数表的进给速度[Vf1 ],求取实际加工时的进给速度[Vf]。
  Obtain Feed (Vf) for actual machining by dividing Recommended Feed (Vf1)from Rate(α).

$$Vf1\,(mm/min) \times \alpha = Vf\,(mm/min)$$

实际加工时的进给速度
Feed for actual machining